

Use-Case Flow

Primary Actor: Compliance Officer

Secondary Actor: Security Auditor (consumer of the export, see extension UC-04)

Preconditions

- The Compliance Officer is authenticated and holds a role authorized to generate exports.
- At least one masked audit log record (produced via UC-01 Ingest Audit Log « include» UC-02 Mask PII Fields) exists in storage for the selected date range.
- The audit log storage service and hash-chain service are online and reachable.

Postconditions

- Success: An immutable export file exists containing the selected audit records, each linked by a SHA-256 hash chain, and the export event itself has been logged as a new audit record (per FR-005).
- Failure: No export file is produced, and the system state (stored audit records, existing hash chain) is unchanged.

Main Success Scenario

1. Compliance Officer selects “Generate Export” and specifies a date range and optional filters (source application, event type).
2. System retrieves all masked audit log records matching the specified criteria from storage.
3. System orders the retrieved records chronologically and computes a SHA-256 hash for each record.
4. System chains each record's hash to the hash of the preceding record, forming a verifiable hash chain for the export set.
5. System packages the records and the hash chain into an export file and digitally signs the file.
6. System writes an audit event recording the export action (actor, timestamp, record count, export ID), itself included in the ongoing hash chain.
7. System presents the Compliance Officer with a confirmation message and a download link for the export file.
8. Use case ends successfully.

Alternate Flow

4a. No Records Match the Selected Criteria

- System finds zero records matching the specified date range/filters after step 2.
- System displays a “no matching records – export not generated” message to the Compliance Officer.
- Compliance Officer may revise the filter criteria and resubmit, or cancel the operation.
- Use case ends without producing an export file (postcondition: failure).

Related Extension

UC-04 Verify Export Integrity « extend» UC-03: after an export is generated, a Security Auditor may optionally invoke integrity verification, re-computing the hash chain of the exported file and comparing it against the stored values to confirm no tampering has occurred.